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Rangkuman Minggu 10 Mikroprosesor dan Interface

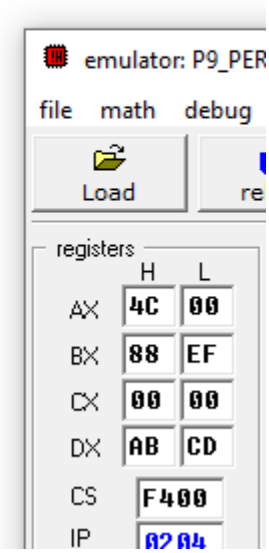
```
TITLE P9_PERKALIAN_8BIT_CR4 ;INPUT DH:DL DAN OUTPUT BX
.MODEL SMALL
.STACK 100H
.DATA
.CODE
MAIN PROC
     CMP DH,0
     JZ AKH
     CMP DL,0
     JZ AKH

     MOV CX,8
     MOV AL,DL
     MOV BX,0

L1:  SHL BX,1
     SHL AL,1
     JNC L0
     ADD BL,DH
     ADC BH,0

L0:  LOOP L1

AKH:
     MOV AH,4CH
     INT 21H
MAIN    ENDP
END MAIN
```

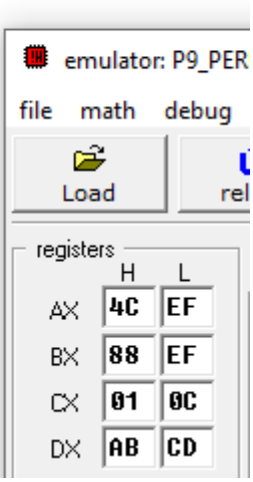


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```
TITLE P9_PERKALIAN_8BIT_CR5 ;INPUT DH:DL DAN OUTPUT BX
.MODEL SMALL
.STACK 100H
.DATA
.CODE
MAIN    PROC
        MOV AL,DL ;INPUT DL
        MOV BL,DH ;INPUT DH
        MUL BL

        ;HASIL AX
        MOV BX,AX

        MOV AH,4CH
        INT 21H
MAIN    ENDP
END MAIN
```



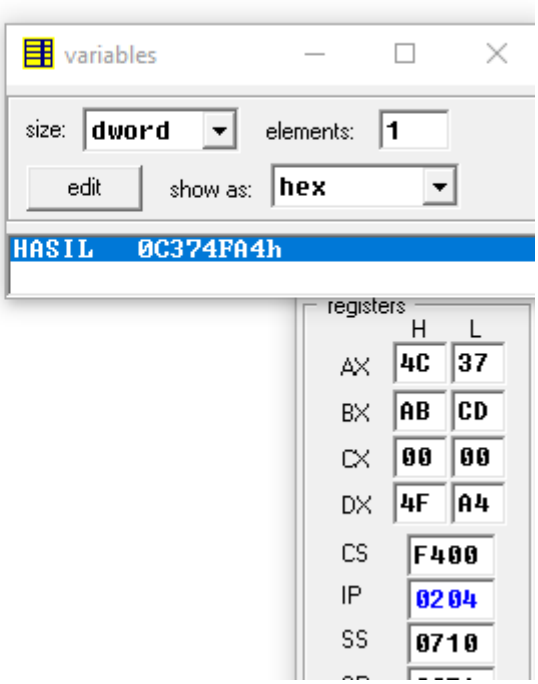
```
TITLE P9_PERKALIAN_16BIT_CR1
;INPUT BX:CX DAN OUTPUT DATA AREA 'HASIL'
.MODEL SMALL
.STACK 100H
.DATA
HASIL DD 0
.CODE
MAIN PROC
     MOV AX,@DATA
     MOV DS,AX
     CMP BX,0
     JZ AKH1
     CMP CX,0
     JZ AKH2

     MOV AX,0
     MOV DX,0

L0:  ADD DX,BX
     ADC AX,0
     LOOP L0

AKH1:
AKH2:
     MOV [HASIL],DX
     MOV [HASIL+2],AX

     MOV AH,4CH
     INT 21H
MAIN ENDP
END MAIN
```



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```
TITLE P9_PERKALIAN_16BIT_CR2
;INPUT BX:CX DAN OUTPUT DATA AREA 'HASIL'
.MODEL SMALL
.STACK 100H
.DATA
HASIL DD 0
TEMP DD 0,0,0,0,0,0,0,0,0,0
.CODE
MAIN    PROC
                MOV AX,@DATA
                MOV DS,AX
                CMP BX,0
                JZ AKH
                CMP CX,0
                JZ AKH

                MOV AX,CX
                MOV AH,AL
                MOV SI,CX
                MOV CX,0
                MOV DX,0

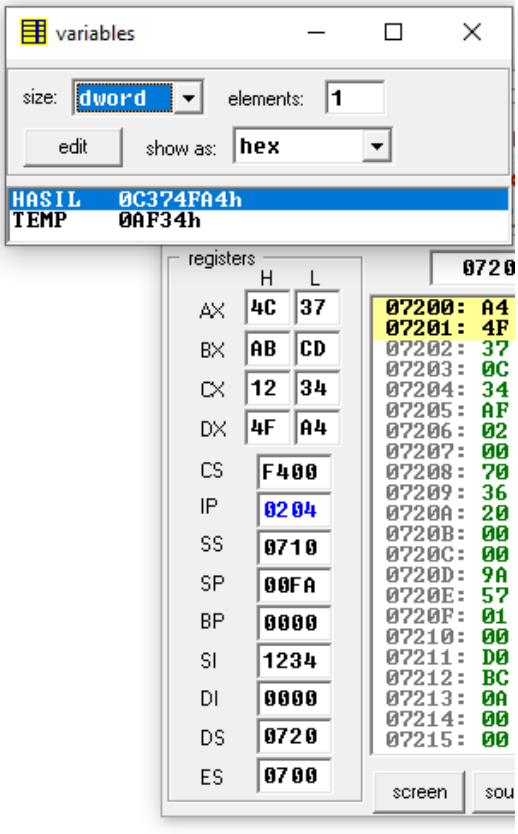
                AND AL,0FH
                ;CMP AL,0
                JZ SKIP1

L0:          DEC AL
                ADD DX,BX
                ADC CX,0
                CMP AL,0
                JNZ L0

SKIP1:  MOV [TEMP],DX
                MOV [TEMP+2],CX

                MOV DX,0
                MOV CX,0

                AND AH,0F0H
                SHR AH,1
                SHR AH,1
                SHR AH,1
                SHR AH,1
                CMP AH,0
                JZ SKIP2
```



```
L1:          DEC AH
               ADD DX,BX
               ADC CX,0
               CMP AH,0
               JNZ L1

               SHL CX,1
               SHL DX,1
               ADC CX,0
               SHL CX,1
               SHL DX,1
               ADC CX,0
               SHL CX,1
               SHL DX,1
               ADC CX,0
               SHL CX,1
               SHL DX,1
               ADC CX,0

SKIP2:  MOV [TEMP+4],DX
               MOV [TEMP+6],CX

               MOV AX,SI
               MOV AL,AH
               MOV CX,0
               MOV DX,0

               AND AL,0FH
               ;CMP AL,0
               JZ SKIP3

L2:          DEC AL
               ADD DX,BX
               ADC CX,0
               CMP AL,0
               JNZ L2
```

```
SKIP3:  MOV [TEMP+9],DX
                MOV [TEMP+11],CX

                MOV DX,0
                MOV CX,0

                AND AH,0F0H
                SHR AH,1
                SHR AH,1
                SHR AH,1
                SHR AH,1
                CMP AH,0
                JZ SKIP4

L3:           DEC AH
                ADD DX,BX
                ADC CX,0
                CMP AH,0
                JNZ L3

                SHL CX,1
                SHL DX,1
                ADC CX,0
                SHL CX,1
                SHL DX,1
                ADC CX,0
                SHL CX,1
                SHL DX,1
                ADC CX,0
                SHL CX,1
                SHL DX,1
                ADC CX,0
```

```
SKIP4:  MOV [TEMP+13],DX
        MOV [TEMP+15],CX

        MOV AX,0
        MOV CX,SI
        MOV DX,0

        MOV DX,[TEMP]
        MOV AX,[TEMP+2]
        ADD DX,[TEMP+4]
        ADC AX,0
        ADD AX,[TEMP+6]
        ADD DX,[TEMP+8]
        ADC AX,0
        ADD AX,[TEMP+10]
        ADD DX,[TEMP+12]
        ADC AX,0
        ADD AX,[TEMP+14]

        MOV [HASIL],DX
        MOV [HASIL+2],AX
 AKH:
        MOV AH,4CH
        INT 21H
MAIN    ENDP
END MAIN
```

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```
TITLE P9_PERKALIAN_16BIT_CR3
;INPUT BX:CX DAN OUTPUT DATA AREA 'HASIL'
.MODEL SMALL
.STACK 100H
.DATA
HASIL DD 0
.CODE
MAIN    PROC
        MOV AX,@DATA
        MOV DS,AX
        CMP BX,0
        JZ AKH
        CMP CX,0
        JZ AKH

        MOV DX,0
        MOV AX,0
        MOV SI,0
        MOV DI,0

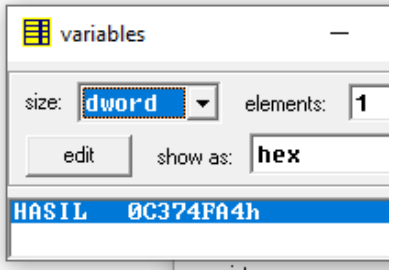
L1:     SHR CX,1
        JNC L0
        MOV DI,AX
        MOV AX,SI
        ADD DX,BX
        ADC AX,DI
        MOV SI,AX
        MOV AX,0
        MOV AX,DI

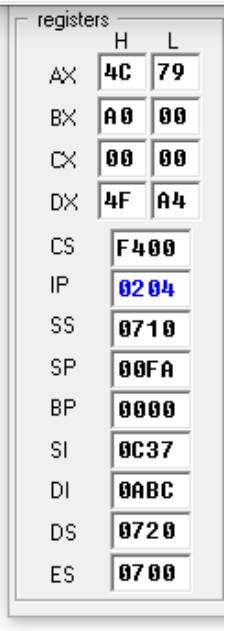
L0:     SHL AX,1
        SHL BX,1
        ADC AX,0

        CMP CX,0
        JNZ L1

AKH:    MOV [HASIL],DX
        MOV [HASIL+2],SI

        MOV AH,4CH
        INT 21H
MAIN ENDP
END MAIN
```





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```
TITLE P9_PERKALIAN_16BIT_CR4
;INPUT BX:CX DAN OUTPUT DATA AREA 'HASIL'
.MODEL SMALL
.STACK 100H
.DATA
HASIL DD 0
.CODE
MAIN    PROC
                MOV AX,@DATA
                MOV DS,AX
                CMP BX,0
                JZ AKH
                CMP CX,0
                JZ AKH

                MOV AX,0
                MOV DX,0
                MOV SI,10H

L1:           SHL AX,1
                SHL DX,1
                ADC AX,0

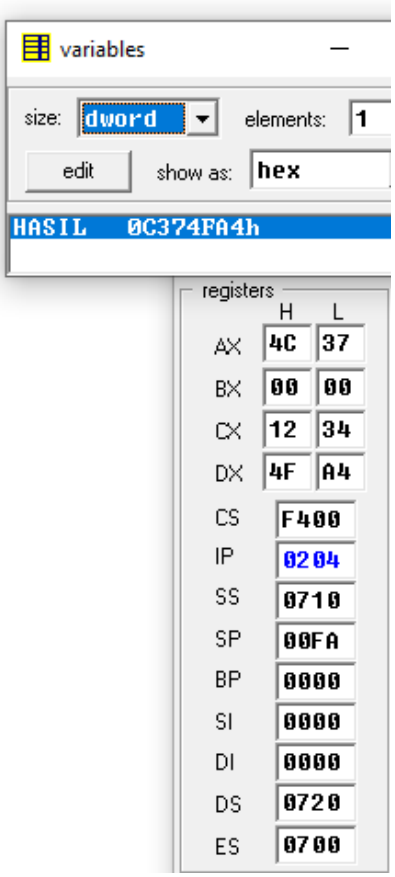
                DEC SI

                SHL BX,1
                JNC L0
                ADD DX,CX
                ADC AX,0

L0:
                CMP SI,0
                JNZ L1

AKH:      MOV [HASIL],DX
                MOV [HASIL+2],AX

                MOV AH,4CH
                INT 21H
MAIN    ENDP
END MAIN
```



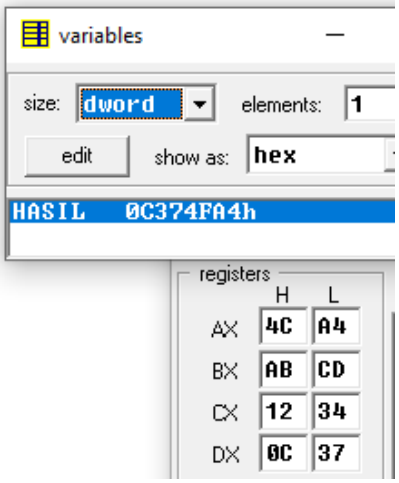
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```
TITLE P9_PERKALIAN_16BIT_CR5
;INPUT BX:CX DAN OUTPUT DATA AREA 'HASIL'
.MODEL SMALL
.STACK 100H
.DATA
HASIL DD 0
.CODE
MAIN PROC
        MOV AX,@DATA
        MOV DS,AX

        ADD BX,0 ;INPUT BX
        ADD CX,0 ;INPUT CX
        MOV AX,CX
        MUL BX

        ;HASIL AX:DX
        MOV [HASIL],AX
        MOV [HASIL+2],DX

        MOV AH,4CH
        INT 21H
MAIN    ENDP
END MAIN
```



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```
TITLE P10_PERKALIAN_24BIT_CR5
;INPUT DATA AREA 'INPUT' DAN OUTPUT DATA AREA 'HASIL'
.MODEL SMALL
.STACK 100H
.DATA
INPUT DD 00AABBCCH,00123456H
TEMP1 DD 0,0,0
TEMP2 DD 0,0,0
HASIL DD 0,0,0
.CODE
MAIN PROC
        MOV AX,@DATA
        MOV DS,AX

        MOV AX,[INPUT]
        MOV BX,[INPUT+4]
        MUL BX
        MOV [TEMP1],AX
        MOV [TEMP1+2],DX

        MOV AX,[INPUT+2]
        MOV DX,0
        MUL BX

        ADD AX,[TEMP1+2]
        ADC DX,0
        MOV [TEMP1+2],AX
        MOV [TEMP1+4],DX

        MOV AX,[INPUT]
        MOV BX,[INPUT+6]
        MUL BX
        MOV [TEMP2],AX
        MOV [TEMP2+2],DX

        MOV AX,[INPUT+2]
        MOV DX,0
        MUL BL
        MOV DX,AX

        ADD DX,[TEMP2+2]
        MOV [TEMP2+2],DX

        MOV AX,0
        MOV BX,0
        MOV CX,0
        MOV DX,0


        MOV AX,[TEMP1+2]
        MOV BX,[TEMP1+4]
        ADD AX,[TEMP2]
        ADC BX,0
        ADD BX,[TEMP2+2]
        ADC CX,0

        ;HASIL
        MOV DX,[TEMP1]
        MOV [HASIL],DX
        MOV [HASIL+2],AX
        MOV [HASIL+4],BX
        MOV [HASIL+6],CX


        MOV AH,4CH
        INT 21H
MAIN    ENDP
END MAIN
```

